

# Smaran Rangarajan Bharadwaj

Bengaluru, Karnataka • +91 89714 46578 • smaranrbharadwaj@gmail.com • linkedin.com/in/smaranrb • github.com/smaranrb

## EDUCATION

- **RV University, Bengaluru** · B.Tech (Hons.), Computer Science and Engineering Graduating: Aug 2026 · CGPA: 9.80
- **Sri Kumaran Children's Home, Bengaluru** · Class 12 (CBSE) - PCMC 2022 · Score: 95.2%

## TECHNICAL SKILLS

**Languages:** Python3, Go, C, C++, Java, JavaScript, YAML  
**Developer Tools:** Docker, Kubernetes, Git, AWS, GCP, Azure, Jira, Confluence, HuggingFace  
**Databases:** MySQL, PostgreSQL, SQLite, MongoDB, ClickHouse  
**Frameworks:** TensorFlow, PyTorch, Node.js, Flask, Streamlit, gRPC, RESTful APIs, Ollama, Kafka, Flink, Unsloth  
**Relevant Coursework:** Google Cloud Computing Foundations, AWS Networking Masterclass, Big Data Computing

## EXPERIENCE

**Software Development Intern** Jan 2026 - Jul 2026  
**Aviatrix Systems Pvt. Ltd.** Bengaluru

- Designed and shipped the scalable Data-Bridge backend in Go using gRPC and ClickHouse materialized views, enabling sub-second queries across 10M+ records.
- Architected and shipped a distributed multi-cloud (AWS/GCP) risk assessment ETL pipeline leveraging Flink and ClickHouse analyzing 22 resource types in under 100 seconds.
- Automated the CI/CD workflows for the cloud-native Data-Bridge service with GitHub Actions, reducing deployment time by 68%.

**Project Intern** Aug 2025 - Jan 2026  
**Aeronautical Development Establishment (DRDO)** Bengaluru

- Architected a multi-stage mission debriefing pipeline processing 10,000+ flight parameters across 6 missions in under 15 minutes with parallelized processing.
- Engineered an event detection engine using Allen's Interval Algebra and expert systems to classify overlapping anomalies.

**ML Engineer Intern** Jun 2025 - Aug 2025  
**GovernAIce** Berkeley (Remote)

- Designed an end-to-end multi-lingual RAG pipeline reducing compliance report generation time by 73% over manual processes by leveraging DeepSeek-R1:70B and MongoDB Atlas Vector Search.
- Developed a policy ingestion pipeline supporting 6 languages and 12,000+ legal documents utilizing IBM Docling and XLM-RoBERTa embeddings.

## LEADERSHIP AND AWARDS

- Founded and established RV University's IEEE Student Branch, growing it from 5 founding ExeCom members to 60+ members across 3 society chapters and 20+ events.
- Awarded merit scholarships for 3 consecutive years, ranking 2nd among the university's top academic performers.
- Won 2nd place among 117 teams at the HPCC Safe Havens Hackathon for developing a distributed computing solution for refugee support systems.
- Co-designed and published a patent for Gelos, a fitness-focused interactive game, alongside Dr. Phani Kumar Pullela.

## PROJECTS

**MITM Attack Simulator** Network Security, Flask, PostgreSQL

- Engineered a full-stack network security toolkit processing 15,000+ packets per second, using Scapy and Flask to simulate MITM attacks across a multi-VM bridged network.
- Secured API endpoints and session audit logging across 3 user roles, by architecting a PostgreSQL database with JWT authentication.

**Obstacle Avoidance for Visually Impaired** Computer Vision, YOLO, Transfer Learning

- Led a four-member team to develop a YOLOv8n-based obstacle detection system for the visually impaired, achieving 82.4% mAP and 132ms average inference latency.

**Braille GPT** Multi-agent System, LoRA

- Developed a multi-agent system built on LoRA-fine-tuned LLaMA 3.1 8B with a Streamlit-based front end, converting text and image inputs into bilingual Braille with a 0.76 average BLEU score across translation agents.